ETC.: OTHER PYQ TYPES

Relations, Venn, Counting, Calendar, Clock, Signs, Letters, Numbers and Rules

OAVS TGT | Reasoning | Topic 19: the “etc.” of the syllabus line

Syllabus line (Reasoning, 30 Marks): “… Figure Matrix and Paper Folding, **etc.**”

Built from: OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). The “etc.” is taken to mean the question types that actually occur in these papers but are **not named in the syllabus line** and do not belong to Topics 1 to 18. Only types with at least one PYQ are covered: **15 types, 23 PYQs**, all solved. The practice questions were generated and checked by program.

What is inside

Ch.	Title	What you get
1	PYQ Analysis	All 23 PYQs grouped into 15 types, priority list
2	Foundation	One rule table: formula and tip for each type
3	Solved PYQs	All 23 PYQs solved step by step (figures included)
4	Shortcuts, Traps and Scope	Speed methods, mistakes, what is left to other topics
5	Practice Set	12 original questions with a verified key
6	Quick Revision	One-page cheat sheet

CHAPTER 1

PYQ Analysis

23 PYQs, 15 types

Type	PYQs	Count
1 Blood relations (coded)	Maths 2024 Q94; Science 2024 Q92	2
2 Venn diagram (data)	SS 2023 Q12	1
3 Embedded / not embedded figures	SS 2023 Q18, Q22	2
4 Counting squares	Eng 2022 Q32	1
5 Cube painted and cut	Eng 2022 Q40	1
6 Calendar	Eng 2022 Q36	1
7 Clock and direction	Eng 2019 Q57	1
8 Mathematical signs	Eng 2022 Q57	1
9 Word letter shift (vowel count)	Maths 2024 Q71, Q98; Science 2024 Q77	3
10 Number digit manipulation	Science 2024 Q81, Q94	2
11 Symbol series counting	Maths 2024 Q83	1
12 Alphabet position	Eng 2019 Q52	1
13 Course of action	Maths 2024 Q97	1
14 Eligibility / criteria	Eng 2019 Q45; Eng 2022 Q50	2
15 Arithmetic reasoning (age, cuts, ordering)	Eng 2019 Q41, Q32; SS 2023 Q23	3

PYQ INSIGHT: Every paper carries 4 or 5 of these. The most repeated types are **letter shift** (3 PYQs across Maths 2024 and Science 2024), **blood relations**, **number manipulation**, **embedded figures** and **eligibility rules** (2 each). All are quick questions (30 to 60 seconds) once the rule is known, so they are the cheapest marks in the section.

Priority	Types
★★★ Must	Letter shift, blood relations, number manipulation, embedded figures, eligibility rules, mathematical signs
★★ Expected	Venn diagram, counting squares, cube painted and cut, calendar, clock, arithmetic reasoning, symbol series
★ Possible	Alphabet position, course of action

CHAPTER 2

Foundation

One rule per type

Type	Rule or formula	Tip
Blood relations (coded)	Translate each symbol, chain left to right, then build the family tree	Use gender clues; “mother’s brother” is the maternal uncle
Venn diagram	Mark which shapes each region lies in; “also / and” means overlap; add the regions	Count each region once
Embedded figure	Pick a distinctive part (circle, arrow, corner) and find it, same size and same direction	No rotation allowed
Counting squares	n × n grid: $1^2 + 2^2 + \ldots + n^2$; add squares formed by diagonals	Count by size
Cube painted and cut (n per edge)	3 faces: 8; 2 faces: $12(n - 2)$; 1 face: $6(n - 2)^2$; 0 faces: $(n - 2)^3$	n = cube root of the number of small cubes
Calendar	Ordinary year = 1 odd day; leap year = 2; odd days = total days ÷ 7 remainder	Count the leap days between the dates
Clock and direction	Hour hand turns 30° per hour; 12 → 3 → 6 → 9 are N → E → S → W when 12 is North	Turn the whole dial
Mathematical signs	Place “=” first, try × and ÷ on large numbers, apply BODMAS	Check both sides
Letter shift	A word has a vowel after the shift if it contains Z, D, H, N or T	Test only these five letters
Number manipulation	Compare whole numbers for highest and lowest; then pick the digit position asked	Work left to right
Symbol series counting	For each symbol check the letter or number on **both** sides	Both conditions
Alphabet position	A = 1 ... Z = 26; n-th to the right: p + n; to the left: p − n	Opposite letter: 27 − p
Course of action	It follows only if it helps solve the stated problem	Reject extreme or unrelated steps
Eligibility / criteria	Check every condition in order; relaxations apply only to the exact case named	“Enrolled” is not “holder”
Arithmetic reasoning	Cuts = pieces − 1; ages differ by a constant; write all weights in one variable	Define one unknown

CHAPTER 3

Solved PYQs

All 23 PYQs

3.1 Blood relations (coded)

Mathematics 2024 Q94

Symbols: + son of, − brother of, × wife of, ÷ father of. Which means U is the mother's brother's son of Z? (A. U + V − W × Y ÷ Z)

1. U + V: U is the son of V. V − W: V is the brother of W. W × Y: W is the wife of Y. Y ÷ Z: Y is the father of Z.

2. W is Z's mother (wife of Z's father). V is W's brother, so V is Z's mother's brother. U is V's son.

Answer: Option A

Science 2024 Q92

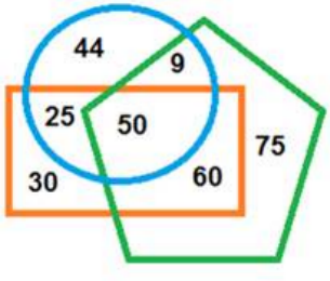
Symbols: + son of, # sister of, ÷ wife of, @ father of. How is P related to K if P + R ÷ G @ H # K?

1. P is the son of R; R is the wife of G; G is the father of H; H is the sister of K.

2. R and G are P's parents, and G is also the father of H and K. So P, H and K are children of the same man: P is the **brother** of K.

Answer: Option 4 (Brother)

3.2 Venn diagram



Social Studies 2023 Q12

Rectangle = Doctors, circle = Females, pentagon = Tax-payers. How many doctors are tax-payers also?

1. Doctors who are tax-payers lie inside both the rectangle and the pentagon.

2. Those regions hold 50 (also in the circle) and 60 (outside the circle): 50 + 60 = 110.

Answer: Option 2 (110)

3.3 Embedded and not-embedded figures



Social Studies 2023 Q18

Select the option in which the given figure is embedded (rotation not allowed).

1. Choose a distinctive part of the given figure: the small circle on a stem with a square on each side.

2. Check each option for that exact shape in the same direction. Only option 4 contains it unchanged (the official key is Option 4).

Answer: Option 4



1 2 3 4 ✓



Social Studies 2023 Q22

Select the option that is NOT embedded in the given figure (rotation not allowed).

1. Check every option for its main feature inside the large figure, in the same direction.

2. Test each option's main feature in the large figure. Options 1, 2 and 4 each have a matching part; option 3 has none, so it is the one that is not embedded (official key Option 3).

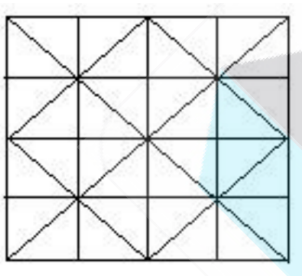
Answer: Option 3



1 2 3 ✓ 4

NOTE: The embedded-figure scans are small, so the reasoning is given as a method; the official keys (Option 4 and Option 3) are as printed in the paper.

3.4 Counting squares and cubes



English 2022 Q32

How many squares are there in the following figure?

1. The figure is a 4 × 4 grid. Grid squares: 16 + 9 + 4 + 1 = 30.
2. The diagonals form small tilted squares (diamonds) with corners at four grid points: 4 of them. One large diamond joins the midpoints of the four sides: 1.
3. Total = 30 + 4 + 1 = 35.
Answer: B. 35

English 2022 Q40

All opposite faces of a big cube are painted yellow, green and purple. The cube is cut into 64 small equal cubes. How many small cubes have three coloured faces?

1. 64 small cubes means 4 per edge. Three painted faces occur only at the 8 corners.
Answer: C. 8

3.5 Calendar and clock

English 2022 Q36

In 1948, 1 February was a Sunday. What was the day on 1 February 1952?

1. From 1 Feb 1948 to 1 Feb 1952 there are 4 years and they include one leap day (29 Feb 1948): 4 × 365 + 1 = 1461 days.
2. 1461 ÷ 7 leaves 5 odd days. Sunday + 5 = Friday.
Answer: C. Friday

English 2019 Q57

The clock shows 3:00. If the minute hand points North, in which direction does the hour hand point?

1. At 3:00 the minute hand is on 12 and the hour hand is on 3, which is 90° clockwise from 12.
2. 90° clockwise from North is East.
Answer: A. East

3.6 Signs, letters and numbers

English 2022 Q57

Select the signs that balance: 72 * 12 * 5 * 35 * 7 * 58 (options: A. ÷, ×, +, −, =).

1. 72 ÷ 12 = 6; 6 × 5 = 30; 30 + 35 = 65; 65 − 7 = 58, and 58 = 58.
Answer: A. ÷, ×, +, −, =

Mathematics 2024 Q71

JET LAG RUM PIN: each letter becomes the next letter of the alphabet. How many new words have no vowel?

1. A shifted word has a vowel only if the old word contains Z, D, H, N or T. JET (T) and PIN (N) do; LAG and RUM do not.
2. Check: KFU (U), MBH, SVN, QJO (O): two words, MBH and SVN, have no vowel.
Answer: B. Two

Mathematics 2024 Q98

WON SEW HAT CUB: same question.

1. WON (N) and HAT (H, T) contain a trigger letter; SEW and CUB do not. So two words (TFX and DVC) have no vowel.
Answer: C. Two

Science 2024 Q77
TRIP SPIN PINK TASK: same question.
1. TRIP (T), SPIN (N), PINK (N) and TASK (T) all contain a trigger letter, so every shifted word has a vowel: USJQ, TQJO, QJOL, UBTL.
Answer: Option 3. Zero

Science 2024 Q81
Numbers: 511 484 496 608 600. Subtract the third digit of the lowest number from the third digit of the highest number.
1. Highest 608, lowest 484. Third digit of 608 is 8; third digit of 484 is 4. $8 - 4 = 4$.
Answer: Option 4. 4

Science 2024 Q94
Numbers: 497 449 483 417 426. Subtract the third digit of the lowest number from the second digit of the highest number.
1. Highest 497, lowest 417. Second digit of 497 is 9; third digit of 417 is 7. $9 - 7 = 2$.
Answer: Option 3. 2

Mathematics 2024 Q83
A # 1 * + B – D 6 7 K % @ & Y 3. How many symbols are immediately preceded by a letter and immediately followed by a number?
1. # (between A and 1) fits. – is between B and D (letter after), % is between K and @, and the rest have a symbol on one side.
Answer: D. One

English 2019 Q52
What is the 14th letter to the left of the twenty-first letter of the alphabet?
1. The 21st letter is U. $21 - 14 = 7$, the 7th letter is G.
Answer: B. G

3.7 Course of action, criteria and arithmetic reasoning

Mathematics 2024 Q97
Saxena wants to quit his BPO job and start a footwear business. Actions: (i) plan the details beforehand; (ii) arrange money; (iii) quit the idea of business.
1. (i) and (ii) help him reach his stated goal. (iii) contradicts it. So both (i) and (ii) follow.
Answer: A. Both (i) and (ii)

English 2019 Q45
Criteria: commerce graduate with at least 60%; age 24 to 40; at least 2 years' experience; at least 55% in selection. Sumitra: age fits; graduate 61%, post-graduate 63% in commerce; 4.5 years' experience; 58.5% in selection.
1. Graduation 61% ≥ 60 ✓. Age ✓. Experience 4.5 ≥ 2 ✓. Selection 58.5 ≥ 55 ✓. All conditions are met, so she is selected.
Answer: A. The candidate is to be selected

English 2022 Q50
Criteria: master's degree; 55% marks (50% for reserved category); 5% relaxation for Ph.D. holders. (I) X, reserved, 52%, can apply. (II) Y, enrolled in a Ph.D. programme, 54%, can apply.
1. I: reserved category needs 50%; 52% ≥ 50, so X can apply.
2. II: the relaxation is for Ph.D. **degree holders**. "Enrolled" is not "holder", so 54% < 55% and Y cannot apply.
Answer: B. Only conclusion I follows

English 2019 Q41
Vamshi is 16 and was 4 times as old as his brother Raju. How old will Vamshi be when he is twice as old as Raju?
1. Raju is $16 \div 4 = 4$. After x years: $16 + x = 2(4 + x)$, so $x = 8$.
2. Vamshi will be $16 + 8 = 24$.
Answer: A. 24

English 2019 Q32
An artist cuts each roll into 10 pieces and makes 36 cuts a minute. How many rolls does he cut in 26 minutes?
1. Cuts in 26 minutes = $36 \times 26 = 936$. One roll into 10 pieces needs 9 cuts.
2. Rolls = $936 \div 9 = 104$.
Answer: A. 104

Social Studies 2023 Q23

P is half of Q; R is twice S; T is 5 kg less than R; U is less than S; Q is half of S. Which rod is the heaviest?

1. Let S = s. Then R = 2s, Q = s/2, P = s/4, T = 2s − 5, U < s.

2. R = 2s is greater than T (2s − 5) and than s. So R is the heaviest.

Answer: Option 2. R

CHAPTER 4

Shortcuts, Traps and Scope

Speed methods and mistakes

- **Letter shift:** look only for Z, D, H, N, T in each word.
- **Blood relations:** write the chain once, then draw a two-level tree.
- **Cubes:** find n first (cube root), then use the formula.
- **Calendar:** count leap days only; every ordinary year adds 1 day, a leap year adds 2.
- **Signs:** put "=" near the last number first, then test × and ÷.
- **Digit questions:** find the highest and lowest **numbers**, not digits.
- **Time:** 30 to 60 seconds per question.

Trap	How to avoid it
Using the wrong symbol meaning	Write the four symbol meanings at the top before reading the chain.
Counting a Venn region twice	Tick each region as you add it.
Forgetting tilted squares	In a figure with diagonals, count diamonds as well as grid squares.
Counting cube corners wrongly	A cube always has 8 corners, whatever its size.
Leap-year slip	A leap day is 29 February only; check whether it lies between the dates.
Comparing digits instead of numbers	Compare whole numbers: 608 is the highest, not the one with the biggest digit.
Treating a vowel as unchanged	After the shift, A, E, I, O, U may appear or vanish; test the old letters Z, D, H, N, T.
Over-reading criteria	Apply relaxations only to the exact case named.

Scope

Related PYQs / topics	Where
Linear, circular, two-row and scheduling arrangements	Arrangement (earlier topic)
Distance and direction walks; shadow direction	Direction (earlier topic)
Substitution and conditional coding	Coding-Decoding (earlier topic)
Mirror, paper folding, matrix, patterns, dice	Topics 10 to 18

CHAPTER 5

Practice Set

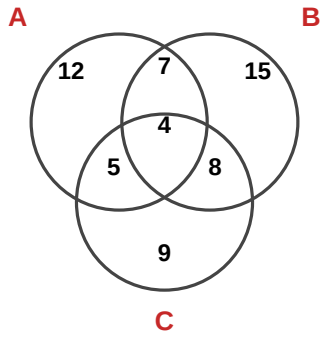
12 original questions; answers on the next page (checked by program)

A. Relations, diagrams and counting

Q1. In a code, ‘A + B’ means A is the son of B; ‘A × B’ means A is the wife of B; ‘A ÷ B’ means A is the father of B; ‘A − B’ means A is the brother of B. In ‘P + Q × R ÷ S’, how is P related to S?

A. Brother B. Cousin C. Uncle D. Father

Q2. In the diagram the numbers show the persons in each section. How many belong to C but not to A?



A. 9 B. 26 C. 17 D. 21

Q3. How many squares of all sizes are there in a 5 × 5 grid (5 small squares along each side)?

A. 25 B. 55 C. 50 D. 30

Q4. A cube painted on all faces is cut into 125 equal small cubes. How many small cubes have exactly two faces painted?

A. 8 B. 54 C. 27 D. 36

B. Calendar and clock

Q5. 15 March 2019 was a Friday. What was the day on 15 March 2023?

A. Wednesday B. Tuesday C. Thursday D. Monday

Q6. The clock shows 9:00. If the hour hand points towards the South, towards which direction does the minute hand point?

A. East B. North C. West D. South-west

Q7. Select the signs that, placed in order in place of *, balance the equation: 8 * 4 * 3 * 2 * 26

A. ×, −, ×, = B. +, ×, −, = C. ÷, +, ×, = D. ×, +, −, =

C. Letters, numbers and rules

Q8. In each of the words SHIN, CLUB, FAST, DRAW every letter is changed to the next letter of the alphabet. How many of the new words have no vowel?

A. None B. One C. Two D. Three

Q9. Numbers (left to right): 632 478 591 346 725. What is the result if the second digit of the lowest number is subtracted from the first digit of the highest number?

A. 2 B. 4 C. 3 D. 5

Q10. Which letter is the 7th to the right of the 9th letter from the left in the English alphabet?

A. O B. Q C. N D. P

Q11. A carpenter cuts a log into 12 equal pieces. Each cut takes 2 minutes. How long does it take?

A. 24 minutes B. 22 minutes C. 20 minutes D. 26 minutes

Q12. Statement: Many students of a school fail in English. Courses of action: (i) the school should arrange extra English classes; (ii) the school should be closed. Which follows?

A. Only (i) B. Only (ii) C. Both (i) and (ii) D. Neither (i) nor (ii)

Answer key

Q	Answer	Reason
1	A	P is the son of Q; Q is the wife of R; R is the father of S. So Q is S's mother and P is her son: P is S's brother (P is male, S's gender is not needed).
2	C	C but not A = C only (9) + B and C only (8) = 17. 26 is all of C; 21 wrongly includes the section common to all three (which lies inside A).
3	B	$1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 1 + 4 + 9 + 16 + 25 = 55$.
4	D	125 small cubes means 5 per edge. Exactly two faces: 12 edges × (5 − 2) = 36 (8 is the corners, 54 is one face, 27 is none).
5	A	4 years from March 2019 to March 2023 contain one leap day (29 Feb 2020): 4 + 1 = 5 odd days. Friday + 5 = Wednesday.
6	C	At 9:00 the minute hand is on 12, which is 90° clockwise from the hour hand at 9. Turning 90° clockwise from South gives West.
7	A	8 × 4 − 3 × 2 = 32 − 6 = 26. The other three options do not balance (checked by program).
8	B	A shifted word has a vowel if the old word contains Z, D, H, N or T (their next letters are A, E, I, O, U). SHIN (H, N), FAST (T), DRAW (D) contain one; only CLUB does not. So one word.
9	C	Highest 725, lowest 346. First digit of 725 is 7; second digit of 346 is 4. 7 − 4 = 3.
10	D	The 9th letter is I; 9 + 7 = 16, and the 16th letter is P.
11	B	12 pieces need 11 cuts (cuts = pieces − 1): 11 × 2 = 22 minutes.
12	A	A course of action follows only if it helps solve the stated problem. Extra classes do; closing the school is an extreme step that does not address the problem.

CHAPTER 6

Quick Revision

One page for the last day

Type	Method
Letter shift	Vowel after shift if the word has Z, D, H, N or T
Cube (n per edge)	8; 12(n − 2); 6(n − 2)²; (n − 2)³
Squares in an n × n grid	$1^2 + 2^2 + \ldots + n^2$ (+ tilted squares)
Calendar	Ordinary 1 odd day, leap 2; remainder ÷ 7
Clock	12 → 3 → 6 → 9 turn 90° clockwise each; rotate the directions
Blood relations	Translate, chain, tree
Cuts	Cuts = pieces − 1
Alphabet	A = 1; right: + n; left: − n; opposite: 27 − p
Criteria	Check all conditions; relaxations only where named

PYQ RULE: Most of these questions need one fact or formula, not a long argument. Learn the 15 rules in the foundation table and you can solve all 23 PYQs in under a minute each.

ROUTINE: Exam-day routine: identify the type from the wording; write the formula; do the single calculation; check the answer against the options. 30 to 60 seconds per question. This completes the syllabus line including “etc.”